Topic 26 — Water Resources & Water Conservation

Covers syllabus (click to expand)

Water Resources | IWRM | Rainwater Harvesting | Watershed Management (वाटरशेड (जलग्रहण) प्रबंधन) | River Conservation | Wetland Conservation | Groundwater Conservation | Drinking Water Quality

Sources baked in: NCERT Geo 11–12; CGWB; CPCB; Jal Shakti; Namami Gange/NMCG; Jal Jeevan Mission; Wetland Rules 2017; IS 10500:2012; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — world water order; CGWB ≠ CPCB; Jal Jeevan year; UP Ramsar traps **Last verified:** August 2026

Current Affairs (this topic)

- Jal Jeevan / Namami Gange (नमामि गंगे) / wetland CA refresh water chapters.
- Keep watershed vs command-area and BOD water-quality traps stable.

26.1 Water Resources

Identity: Oceans dominate | Rivers tiny fraction | India water-stressed | Surface + groundwater

- Earth's water: oceans ~**97.2%** (saline (लवणीय)); polar ice/glaciers ~**2%**; groundwater ~**0.6%**; rivers/lakes ~**0.01%**.
- India holds ~**4%** of world renewable freshwater but ~**18%** of population — per capita availability ~**1,400 m³/year** (stress below **1,700 m³**).
- **Surface water:** rivers, lakes, reservoirs — highly seasonal under monsoon.
- **Groundwater:** India's largest **extractable** freshwater source (~45% irrigation, ~80% rural domestic).
- **Ganga–Brahmaputra (ब्रह्मपुत्र)** system carries about **40%** of India's river water.
- Drivers of stress: population, urbanisation, inefficient irrigation, pollution reducing usable supply.
- The **hydrological / water cycle** moves water through evaporation, condensation, precipitation, infiltration, and runoff.
- **Solar radiation** plays the main role in driving the water cycle by evaporating water from oceans, lakes, and rivers.
- The water cycle is essentially a **closed system** for planetary water volume on human timescales.
- **Humidity** measures water-vapour content of air and is measured with a **hygrometer** or psychrometer.
- Higher humidity means more vapour available for clouds, so **amount of rain** depends strongly on atmospheric humidity when cooling occurs.
- Many transplanted seedlings fail because most **root hairs** are lost during transplantation. Root hairs absorb water and mineral nutrients from soil.
- **Biotic resources** come from living or organic material (forests, animals, fish) and from materials obtained from them. **Coal and petroleum** are counted as biotic because they form from decayed

organic matter.

- **Freshwater / pure water is not a biotic resource** — it is an abiotic natural resource, and usable drinking freshwater is scarce, so conservation techniques matter.
- **World Water Day (22 March)** highlights freshwater and sustainable management (UN designation; first observed **1993**; **UN-Water** coordination from **2003**). Theme **2024**: *Leveraging Water for Peace*.
- **Rajendra Singh** of Alwar (Rajasthan), called **Jal Purush / Waterman of India**, runs **Tarun Bharat Sangh** for community water harvesting; awards include **Ramon Magsaysay (2001)** and **Stockholm Water Prize (2015)**.
- The **Lisbon Declaration** (*Our Ocean, Our Future, Our Responsibility*) from the UN Ocean Conference is about **ocean** conservation and marine pollution — not freshwater glaciers.

PYQ — UPPCS Prelims 2025, Q88

Arrange descending: (1) Oceans (2) Rivers (3) Polar ice and glaciers (4) Groundwater

Options: A. 1, 3, 4, 2 B. 1, 2, 4, 3 C. 1, 2, 3, 4 D. 1, 4, 3, 2

Show answer

Ans: A — Oceans > Ice > Groundwater > Rivers.

> **Logic:** Rivers are the **smallest** fraction — not second after oceans. Freshwater ≠ biotic.

26.2 Integrated Water Resources Management (IWRM) (एकीकृत जल संसाधन प्रबंधन)

Identity: Dublin 1992 | Basin unit | National Water Policy 2012 | Jal Shakti 2019

- **IWRM** coordinates water, land, and related resources within a **river basin** to balance economy, society, and ecosystems.
- **Dublin Principles (1992):** freshwater is finite; participatory management; women central; water has **economic value**.
- **National Water Policy 2012:** river basin as planning unit; allocation priority **drinking water > ecology > agriculture > industry**.
- **Ecological flows (e-flows):** minimum release below dams for river ecology.
- **Conjunctive use:** coordinated surface water + groundwater management.
- **Ministry of Jal Shakti (May 2019)** merged Water Resources with Drinking Water & Sanitation — institutional home for IWRM.
- IWRM ≠ only building dams — includes demand management, quality, ecology, and stakeholder participation.
- **National Water Mission (NWM)** is one of eight **NAPCC** missions (plan launched **30 June 2008**). Goal: conserve water, cut wastage, and ensure more equitable distribution across and within states through integrated water resource development and management.

- NWM's five goals: (1) comprehensive water database in the public domain and assessment of climate-change impacts on water; (2) citizen and state action for conservation, augmentation, and preservation; (3) focused attention to vulnerable / over-exploited areas; (4) increasing water-use efficiency by **20%**; (5) promotion of basin-level IWRM.
- Proper NWM implementation supports urban **wastewater recycling** and, for water-scarce coastal cities, technologies that allow use of **ocean water**. It does **not** include linking all Himalayan rivers to peninsular rivers or fully reimbursing farmers' borewell costs.

Traditional	IWRM
Departmental silos	River basin planning
Supply focus	Supply + demand + quality
Environment (पर्यावरण) ignored	Ecological flows included

> **Logic:** "IWRM means maximum irrigation extraction" = **FALSE**. River linking ≠ NWM package.

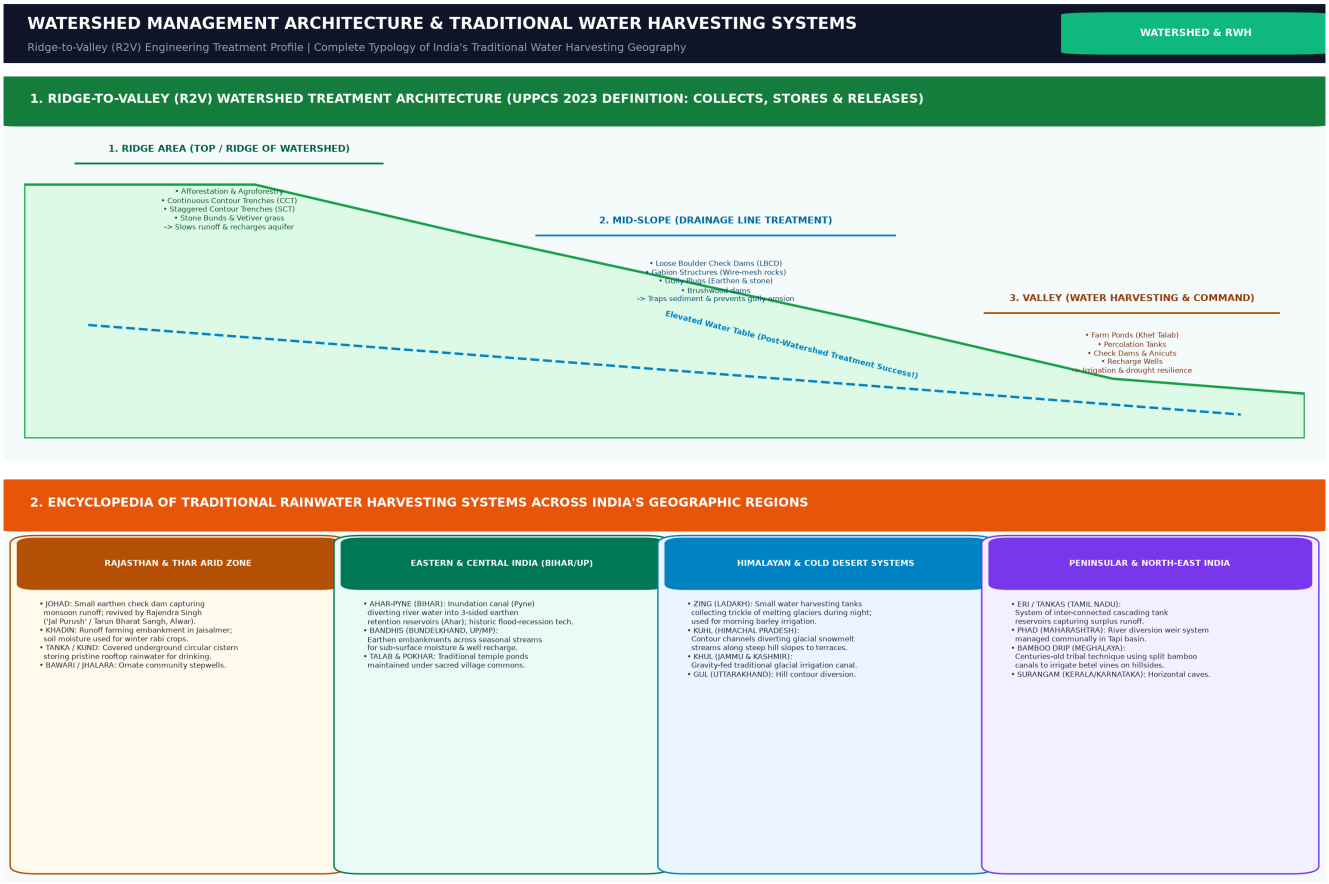
26.3 Rainwater Harvesting

Identity: Capture + store/recharge | Tamil Nadu 2003 mandate | Dual flood + aquifer benefit

- **Rainwater harvesting** collects rooftop or surface runoff for **direct use** or **aquifer recharge** before it drains away.
- **Rooftop:** roof → filter → tank (or recharge pit).
- **Surface:** check dams, percolation tanks, farm ponds, contour trenches.
- **Traditional:** Rajasthan **johad** (community pond) and **taanka** (cistern).
- **Benefits:** reduces urban flood peaks, recharges groundwater, supplements municipal supply.
- **Tamil Nadu (2003)** first made RWH **mandatory** in new buildings — Chennai model widely cited.
- First-flush diversion improves stored water quality; RWH supplements, does not replace, bulk supply.

Method	Function
Rooftop tank	Direct use
Recharge pit/well	Aquifer recharge
Check dam (दाम -)	Slow runoff + percolation
Johad/taanka	Traditional arid storage

> **Logic:** RWH = both storage **and** recharge — not tanks alone.



26.4 Watershed Management

Identity: Ridge-to-valley | Drainage basin | IWMP / Hariyali | Collects-stores-releases water

- A **watershed** is the land area draining to a common outlet — also called catchment/drainage basin.
- It **collects, stores, and releases** water — UPPCS 2023 definition fact.
- **Ridge-to-valley treatment:** afforestation and contour bunds at top → check dams mid-slope → farm ponds at bottom.
- Measures: contour bunding, terracing, nala bunding, gully plugs, percolation tanks.
- **IWMP** (Integrated Watershed Management Programme) — central watershed scheme; converges with MGNREGA.
- **Hariyali** — Government of India watershed project enabling rural water conservation (2022 PYQ).
- Classic success models: **Sukhomajri (Haryana)**, **Ralegan Siddhi (Maharashtra)**.
- Outcomes: less erosion, higher water table, drought (सूखा) resilience, lower flood peaks.
- Desert plants reduce water loss through **hard and waxy leaves**, **tiny or absent leaves**, and **thorns instead of leaves** — all three modifications inhibit water loss.
- Heavy **sand mining** in riverbeds can pollute groundwater and lower the water table (sand acts as a sponge/filter). It does **not** typically decrease river salinity as a listed consequence.

PYQ — UPPCS Prelims 2023, Q110

A geographical unit that collects, stores and releases water is called—

Options: A. Watershed B. Wasteland C. Wetland D. Water Reserve

Show answer

Ans: A — Watershed / drainage basin.

PYQ — UPPCS Prelims 2022

Watershed Development Project enabling rural population to conserve water:

Options: A. Jal Kranti B. Pani Sansad C. Hariyali D. None

Show answer

Ans: C — **Hariyali**.

> **Logic:** Watershed management **reduces** soil erosion — does not increase it.

26.5 River Conservation

Identity: GAP 1985 → NGRBA 2009 → Namami Gange 2014 | YAP | Dolphin | STP/CETP

- River degradation drivers: untreated sewage, industrial effluent, sand mining, floodplain encroachment, flow alteration by dams.
- **Ganga** (गंगा) was declared a **National River** on **4 November 2008**.
- **Central Ganga Authority (Feb 1985)** under Rajiv Gandhi (गांधी) launched the **Ganga Action Plan** (गंगा एक्शन प्लान); renamed **National River Conservation Authority (NRCA)** in **September 1995** (PM as chair).
- **GAP Phase I = 1985**. Phase II (**1993**) brought in **Yamuna Action Plan** and **Gomti Action Plan**.
- **Yamuna Action Plan:** formally launched **1993** (YAP-I ~1993–2003; YAP-II Dec **2004**; **YAP-III from 7 May 2016** under Namami Gange). *Maili Se Nirmal Yamuna* revitalisation framing from **2017**.
- **NGRBA (National Ganga River Basin Authority)** constituted **20 February 2009** under **EPA 1986**. The **Prime Minister** chairs it; CMs of Ganga states are members — chair is **not** rotational among CMs. It plans, finances, monitors, and coordinates basin-scale Ganga conservation.
- **Namami Gange, 2014** (Budget outlay teaching): integrated Ganga rejuvenation — STPs/CETPs, river-front/ghat works, surface cleaning, biodiversity, afforestation, industrial effluent monitoring, Ganga Gram. Pillars include **Nirmal Dhara (धारा)** and **Aviral Dhara**.
- **NMCG (National Mission for Clean Ganga)** under **Jal Shakti** implements Namami Gange.
- **NRCP** works: intercept/divert raw (रॉ) sewage, STPs, low-cost sanitation on banks, electric/improved wood crematoria, public awareness.
- **STP** treats domestic sewage; **CETP** treats combined industrial effluent (e.g. Kanpur tanneries — chromium).
- Clean stretch targets: BOD **<3 mg/L**, DO **>6 mg/L**.
- **UP facts: Kanpur** (कानपुर) = Ganga tannery stretch; **Gomti** = Lucknow municipal sewage; **Varanasi** (वाराणसी) = sewage + ghat waste.

- **Ganges River Dolphin** (*Platanista gangetica*) is one of four (चातुर्याम) obligate freshwater dolphins; declared India's **National Aquatic Animal in 2009**. Decline drivers include dams/barrages, accidental fishing-net capture, and agrochemicals — **not** mainly rising crocodile numbers.
- Jharkhand State Ganga River Conservation Authority was formed in **2009** (CM as ex-officio chair).

Programme	Year	Fact
Central Ganga Authority / GAP	**1985**	First Ganga STPs
NRCA rename	**1995**	Broader river mandate
YAP launch	**1993**	Yamuna bilateral Japan–India track
NGRBA	**2009**	PM-chaired basin authority
Namami Gange	**2014**	NMCG nodal
YAP-III	**2016**	Under Namami Gange
NRCP	1995	Other rivers

PYQ — UPPCS Prelims 2022

Main purpose of Ganga Action Plan (1985):

Options: A. Make water pollution-free B. Irrigation only C. Build new dams D. None

Show answer

Ans: A — Pollution abatement / make water pollution-free.

> **Logic:** Namami Gange = **2014**, not 1985. NGRBA chair = **PM**, not rotating CM. Gomti ≠ Kanpur.

26.6 Wetland Conservation

Identity: Ramsar 1971 | NLCP 2001 → NPCA 2013 | Wetland Rules 2017 | UP Ramsar geography

- Wetlands provide flood absorption, groundwater recharge, purification, fisheries, and bird habitat (वास स्थान).
- **Ramsar Convention (1971) (रामसर कन्वेंशन):** wetland conservation and **wise use** — World Wetlands Day = **2 February**.
- India joined Ramsar **1982**; Wetland Rules **2017** (MoEFCC (पर्यावरण, वन और जलवायु परिवर्तन मंत्रालय)) regulate notified wetlands.
- **National Lake Conservation Plan (NLCP)** launched **June 2001** to restore degraded urban and semi-urban lakes via an integrated ecosystem (पारिस्थितिकी तंत्र) approach (pollution control from wastewater inflows).
- In **February 2013**, CCEA approved merger of NLCP and the **National Wetlands Conservation Programme** into the **National Plan for Conservation of Aquatic Ecosystems (NPCA)**.

- Classic NLCP teaching lakes: **Bhoj (MP)**, **Sukhna (Chandigarh)**, **Pichola (Udaipur (उदयपुर))**, **Bhimtal / Nainital (नैनीताल) set (Uttarakhand (उत्तराखंड))**, Powai, Pushkar, Ooty, Ramgarh Tal (Gorakhpur (गोरखपुर)), Mansi Ganga (Mathura (मथुरा)), Lakshmi Tal (Jhansi (झांसी)). **Chilika (Odisha) is not under NLCP**.
- **UP Ramsar sites** include Sarsai Nawar, Samaspur, Nawabganj, Upper Ganga, Bakhira, Haiderpur.
- **NOT in UP: Rudrasagar Lake (Tripura)** and **Sultanpur NP (Haryana)** — classic state-trap pair.

> **Logic:** Full wetland ecology depth sits in Topic 21 — here fact **conservation schemes + UP geography**. Chilika ≠ NLCP.

26.7 Groundwater Conservation

Identity: CGWB | Atal Bhujal 2019 | Over-extraction belt | Arsenic & fluoride

- Groundwater is India's main **extractable** freshwater stock — aquifers recharged by monsoon infiltration.
- **CGWB (Central Ground Water Board)** under Jal Shakti assesses quantity and quality.
- **Annual Groundwater Quality Report** is prepared by **CGWB** — not CPCB or CWC (केंद्रीय जल आयोग).
- **CGWA (Central Ground Water Authority)** was formed under the **Environment (Protection) Act, 1986** to regulate and control groundwater development — **CGWA ≠ CGWB**.
- India has the **largest area under groundwater irrigation** in the world (followed by China and the USA).
- **Over-extraction hotspots:** Punjab–Haryana–western UP Green Revolution (हरित क्रांति) belt — falling water tables.
- **Contamination: arsenic** in Ganga–Brahmaputra alluvium (urban drinking-water teaching often cites (साइट्स) **Kolkata**); **fluoride** in parts of Rajasthan and UP.
- **Atal Bhujal Yojana (2019):** community-led groundwater management in water-stressed blocks (7 states).
- Conservation tools: recharge structures, crop diversification, regulated tubewell electricity, conjunctive use.
- Legal stack reminder: **Water (Prevention and Control of Pollution) Act, 1974** (amended **1988**); **Water Cess Act, 1977** (enforced **1 April 1978**).

PYQ — UPPCS Prelims 2025, Q70

Annual Groundwater Quality Report 2024 prepared by: (1) CPCB (2) CWC (3) CGWB

Options: A. 1 and 2 B. Only 3 C. 2 and 3 D. Only 1

Show answer

Ans: B — **CGWB only**.

Body	Role
CGWB	Groundwater quantity/quality assessment
CGWA	Groundwater regulation (EPA 1986)
CPCB	Surface water & air pollution
CWC	Surface water/irrigation data

> **Logic:** CGWB ≠ CPCB; CGWA ≠ CGWB.

26.8 Drinking Water Quality

Identity: Jal Jeevan Mission 2019 | Disinfection | IS 10500:2012 | SDG 6

- **Jal Jeevan Mission (JJM):** announced **August 2019** / Union Budget **FY 2019–20** — functional household tap connection (**Har Ghar Jal**) in rural India.
- Nodal: Ministry of Jal Shakti; aligns with **SDG 6** (Clean Water and Sanitation).
- **IS 10500:2012 (BIS)** drinking water standard: pH **6.5–8.5**; fluoride **≤1.5 mg/L**; arsenic **≤0.01 mg/L**; faecal coliform ideally **zero**.
- Disinfection / purification tools: **chlorine** gas and bleaching powder (classic primary bactericidal step); also **chloramine (NH_2Cl)**, **chlorine dioxide (ClO_2)**, **ozone**, iodine compounds, heat, and **ultraviolet radiation**.
- **UV purification** inactivates/kills harmful microorganisms without adding chemicals and without changing taste, smell, or colour. UV does **not** remove all undesirable odours or quicken sedimentation/turbidity clearance — those claims are false in UPSC framing.
- Quality risks: microbial contamination from sewage; fluoride/arsenic geogenic contamination (Kolkata arsenic teaching); industrial toxins near polluted stretches.
- Swachh Bharat (2014) + JJM (2019) together address sanitation and drinking water access.

PYQ — UPPCS Prelims 2022, Q145

Jal Jeevan Mission was declared in Union Budget of which financial year?

Options: A. 2021–22 B. 2022–23 C. 2020–21 D. 2019–20

Show answer

Ans: D — **2019–20**.

Parameter	Limit (IS 10500)
pH	6.5–8.5
Fluoride	≤1.5 mg/L
Arsenic	≤0.01 mg/L

> **Logic:** JJM = **2019**, not 2014 (Swachh Bharat year). UV = microbial kill only among the three classic claims.
